

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T55 · Cumulative Frequency

With Answers

Questions with full worked solutions

■ 64 pages

FAMILY MAP

Cumulative Frequency Diagrams

F01	Construct cumulative-frequency diagrams	5 Q 18 marks
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F02	Read counts and thresholds from cumulative frequency	9 Q 42 marks
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F03	Estimate and compare location and spread from cumulative frequency	9 Q 19 marks
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- 13 The frequency table gives information about the times, in minutes, that 60 people took to complete a puzzle.

Time (t minutes)	Frequency
$0 < t \leq 10$	8
$10 < t \leq 20$	13
$20 < t \leq 30$	12
$30 < t \leq 40$	17
$40 < t \leq 50$	7
$50 < t \leq 60$	3

- (a) Complete the cumulative frequency table.

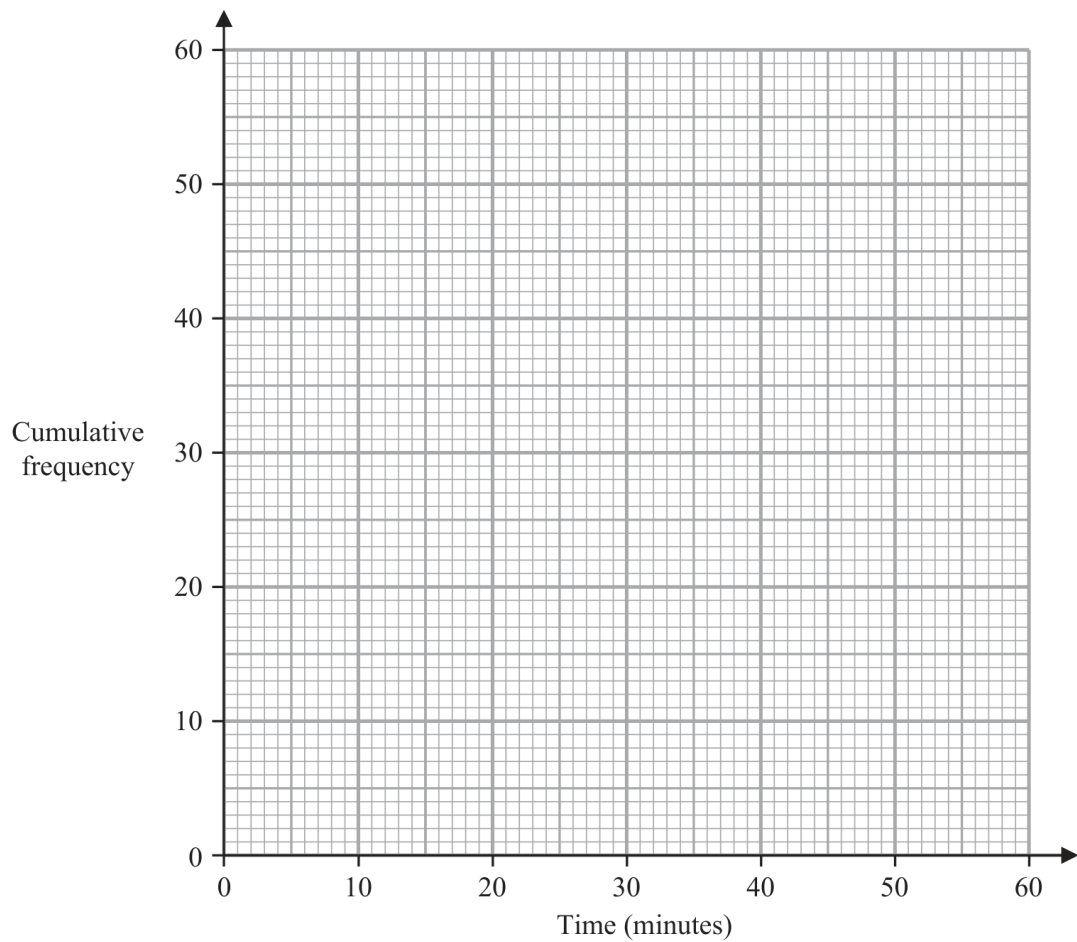
Time (t minutes)	Cumulative frequency
$0 < t \leq 10$	
$0 < t \leq 20$	
$0 < t \leq 30$	
$0 < t \leq 40$	
$0 < t \leq 50$	
$0 < t \leq 60$	

(1)

- (b) On the grid opposite, draw a cumulative frequency graph for your table.

Continued
4MA1-2H Q13

ORIGINAL QUESTION



(2)

- (c) Use your graph to find an estimate for the median time.

..... minutes
(1)

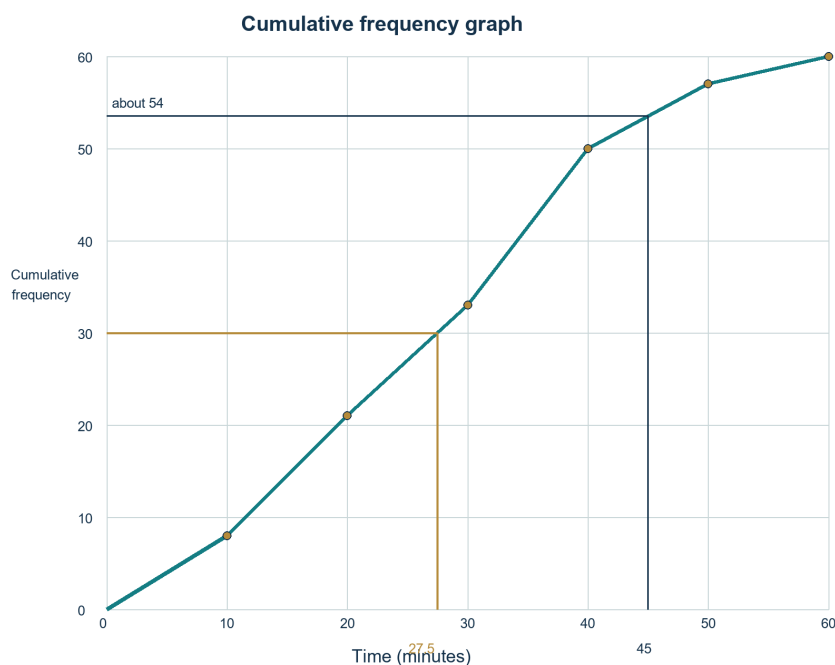
- (d) Use your graph to find an estimate for the number of these people who took more than 45 minutes to complete the puzzle.

.....
(2)

(Total for Question 13 is 6 marks)

Worked Solution 26

Cumulative frequency graph and estimates

UNIT 2
6 marks**Cumulative totals** $8, 8 + 13 = 21, 33, 50, 57, 60.$ **Plot** $(10, 8), (20, 21), (30, 33), (40, 50), (50, 57), (60, 60)$, joined as a cumulative-frequency graph.**Read estimates**Median: 27 to 28 minutes; more than 45 minutes: $60 - 54 \approx 6$ people.**Final answer** $8, 21, 33, 50, 57, 60$; median 27 to 28 min; about 6 people

Question 18

4MA1-2H Q13

ORIGINAL QUESTION

7 marks

13 The table gives information about the distances, in km, that 70 teachers travel to school.

Distance (d km)	Frequency
$0 < d \leq 10$	7
$10 < d \leq 20$	17
$20 < d \leq 30$	18
$30 < d \leq 40$	14
$40 < d \leq 50$	10
$50 < d \leq 60$	4

(a) Complete the cumulative frequency table.

Distance (d km)	Cumulative frequency
$0 < d \leq 10$	
$0 < d \leq 20$	
$0 < d \leq 30$	
$0 < d \leq 40$	
$0 < d \leq 50$	
$0 < d \leq 60$	

(1)

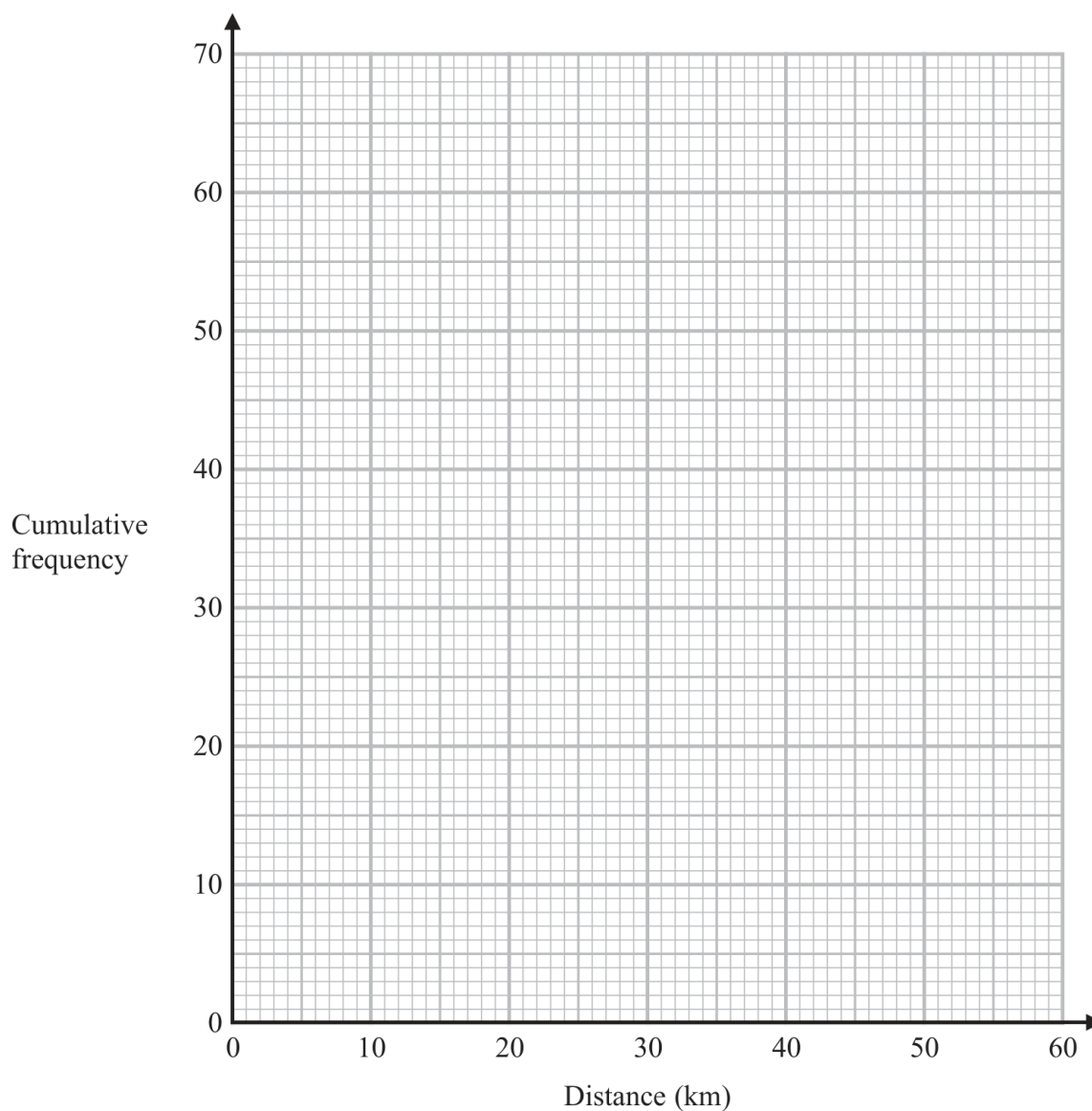
Question 18 (continued)

4MA1-2H Q13

ORIGINAL QUESTION**7 marks**

(b) On the grid opposite, draw a cumulative frequency graph for your table.

(2)



(c) Use your graph to find an estimate for the interquartile range of the distances.

..... km

(2)

Question 18 (continued)

4MA1-2H Q13

ORIGINAL QUESTION

7 marks

- (d) Use your graph to find an estimate for the number of teachers who travel more than 46 km.

.....

(2)

Worked Solution 18

Cumulative frequency

MODULAR UNIT 2

7 marks

Table $7, 24, 42, 56, 66, 70.$ **Graph**Plot $(10, 7), (20, 24), (30, 42), (40, 56), (50, 66), (60, 70)$ and join smoothly.**IQR**

$$Q_1 \approx 16.2, Q_3 \approx 37.5 \Rightarrow \text{IQR} \approx 21.3 \text{ km}.$$

More than 46 km

$$70 - \text{CF}(46) \approx 70 - 62 = 8.$$

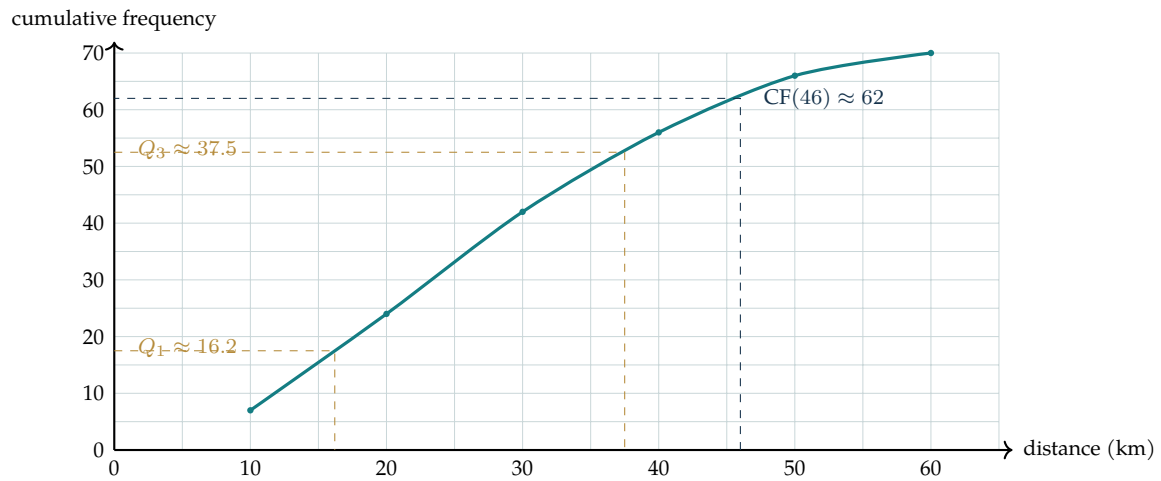
Final answerCF 7, 24, 42, 56, 66, 70; IQR ≈ 21.3 km; 8 teachers

Completed diagram 18

Cumulative frequency

MODULAR UNIT 2

7 marks

Completed solution diagram: Cumulative-frequency graph

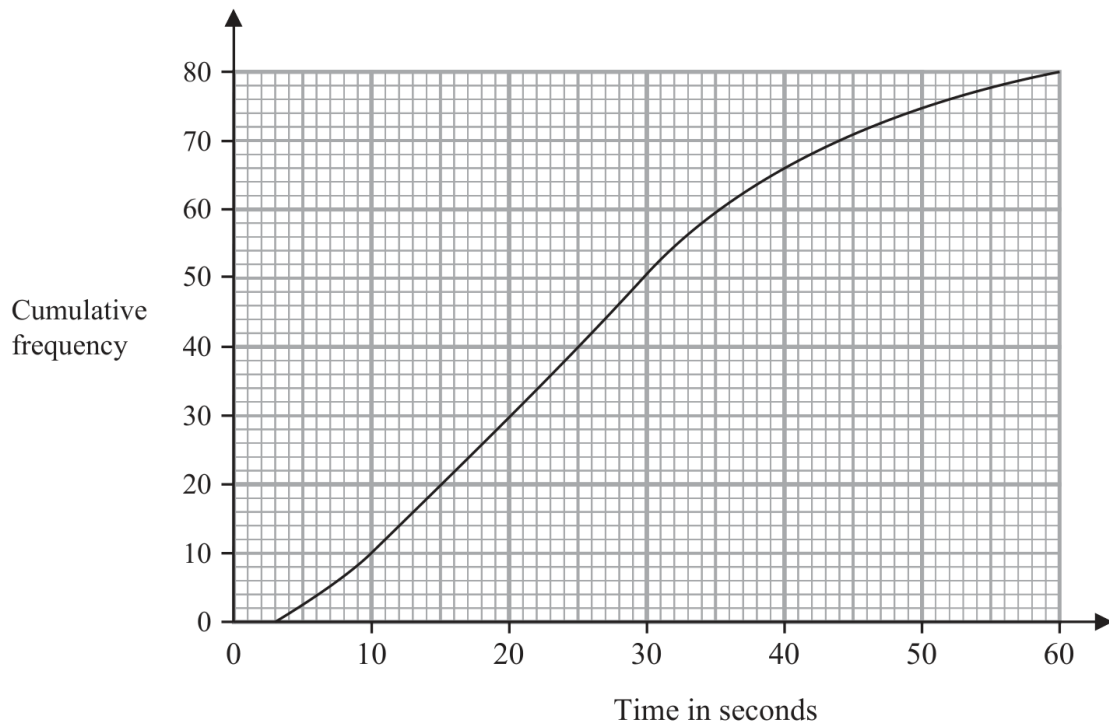
Question 09

4MA1-1H Q13 (a)

ORIGINAL QUESTION UNIT 2

1 marks

- 13 The cumulative frequency graph gives information about the times, in seconds, that 80 adults took to log in to an online bank.



- (a) Find an estimate for the median time.

..... seconds
(1)

Worked solution 09

Median from cumulative frequency

MODULAR UNIT 2

1 marks

(a) – Median from cumulative frequency

Read the middle rank

 $80/2 = 40$; reading $CF = 40$ from the graph gives about 25 seconds.**Final answer**

25 seconds

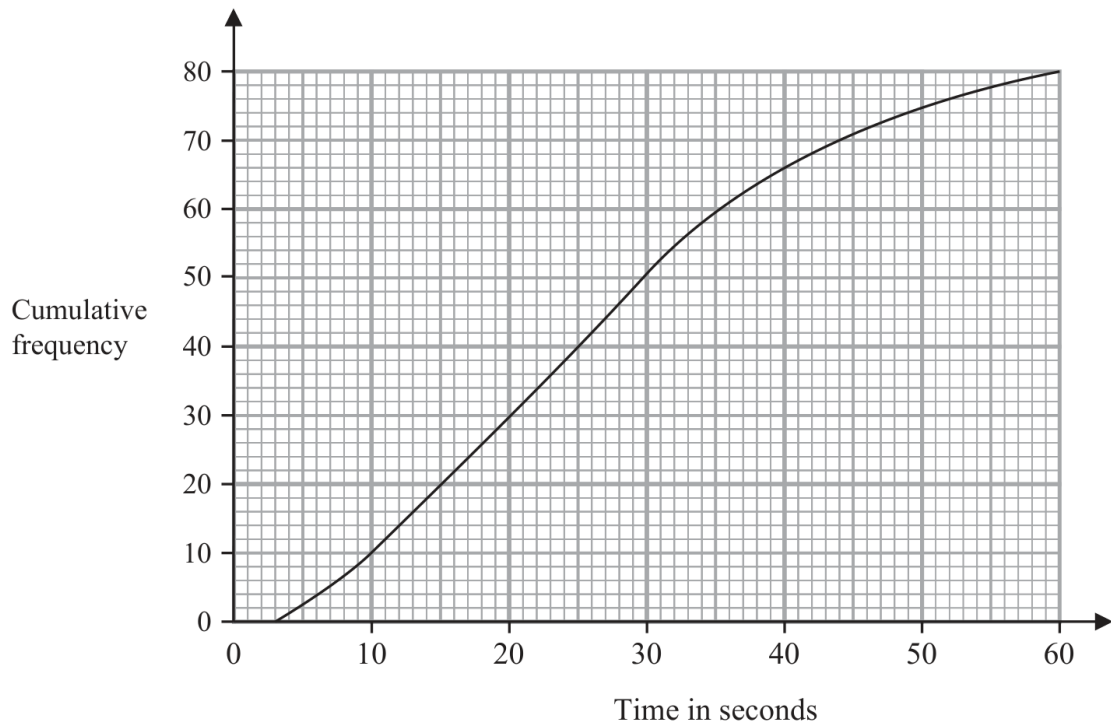
Question 10

4MA1-1H Q13 (b)

ORIGINAL QUESTION UNIT 2

3 marks

- 13 The cumulative frequency graph gives information about the times, in seconds, that 80 adults took to log in to an online bank.



- (b) Work out the percentage of these adults that took longer than 50 seconds to log in.
Show your working clearly.

..... %
(3)

Worked solution 10

Percentage taking longer than 50 seconds

MODULAR UNIT 2

3 marks

(b) – Percentage taking longer than 50 seconds

Read CF at 50

$$CF(50) \approx 75, \quad 80 - 75 = 5.$$

Convert to a percentage

$$\frac{5}{80} \times 100 = 6.25\%.$$

Final answer

6.25%

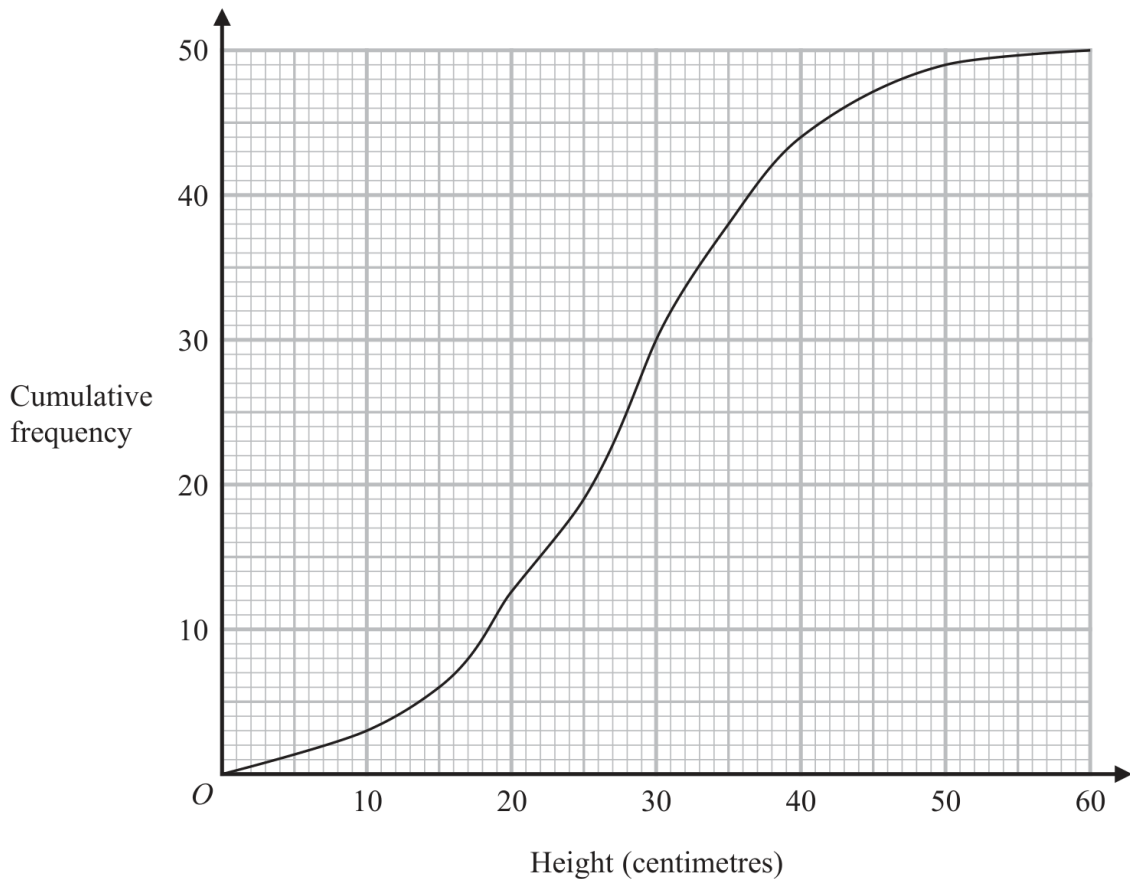
Question 12

4MA1-1H Q12

ORIGINAL QUESTION

4 marks

- 12** The cumulative frequency graph shows information about the heights, in centimetres, of 50 plants in a flowerbed.



- (a) Use the graph to find an estimate for the median height of these plants.

..... centimetres
(1)

- (b) Use the graph to find the frequency for the class interval $30 < \text{Height} \leq 40$

.....
(1)

- (c) Use the graph to find an estimate for the number of plants with a height greater than 35 centimetres.

.....
(2)

Worked Solution 12

Read a cumulative-frequency graph

MODULAR UNIT 2

4 marks

Median

$$CF = 25 \Rightarrow h \approx 28.$$

Class frequency

$$CF(40) - CF(30) \approx 44 - 30 = 14.$$

Above 35 cm

$$50 - CF(35) \approx 50 - 38 = 12.$$

Final answer

$$28 \text{ cm}; \quad 14; \quad 12$$

Original question 27

4MA1-2H Q13 M01, M02, M03, M04

MODULAR UNIT 2

6 marks

- 13 The frequency table gives information about the weights, in kilograms, of 60 parcels in a delivery van.

Weight (w kilograms)	Frequency
$0 < w \leq 1$	4
$1 < w \leq 2$	15
$2 < w \leq 3$	20
$3 < w \leq 4$	11
$4 < w \leq 5$	6
$5 < w \leq 6$	4

- (a) Complete the cumulative frequency table.

Weight (w kilograms)	Cumulative frequency
$0 < w \leq 1$	
$0 < w \leq 2$	
$0 < w \leq 3$	
$0 < w \leq 4$	
$0 < w \leq 5$	
$0 < w \leq 6$	

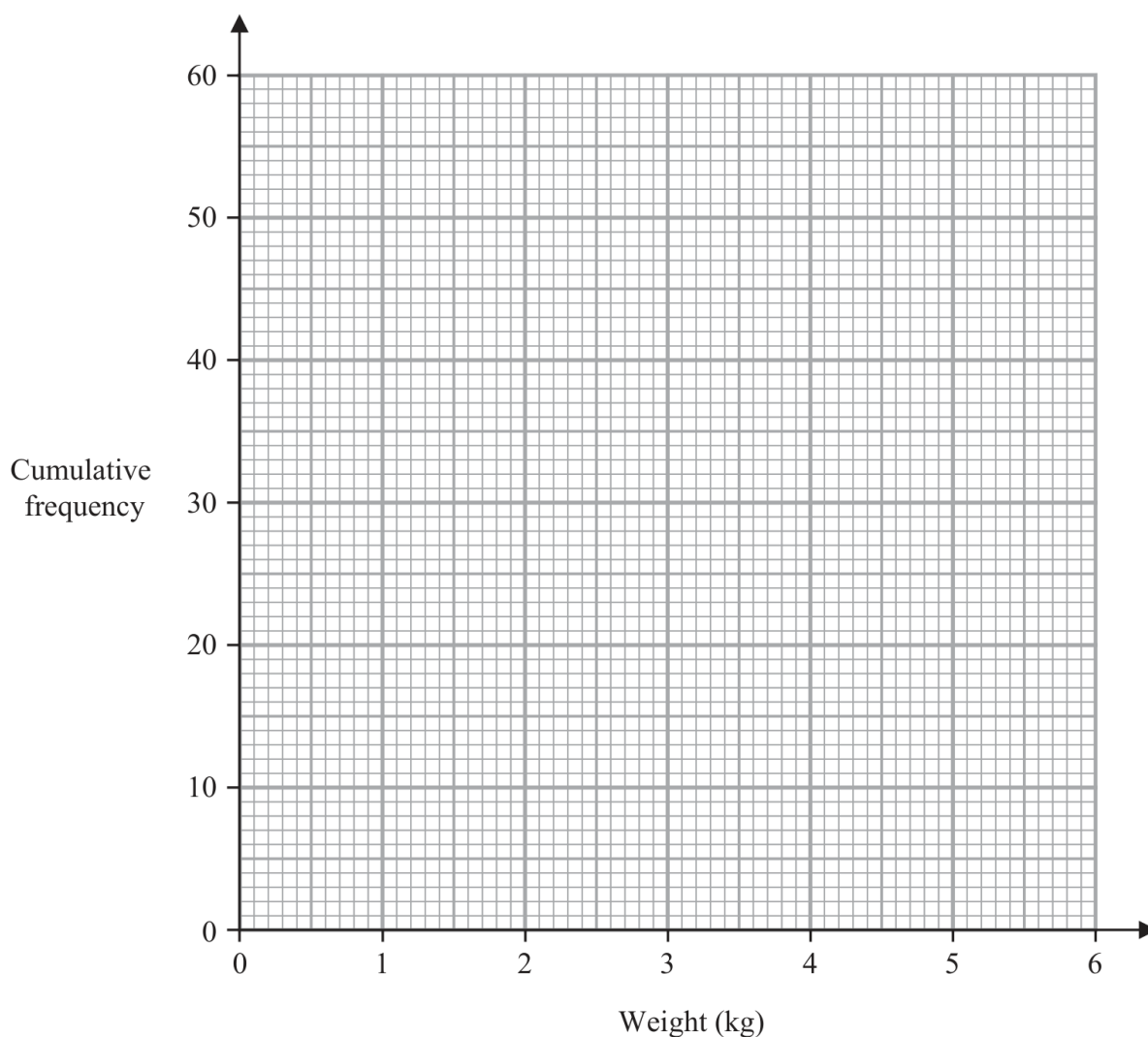
(1)

- (b) On the grid opposite, draw a cumulative frequency graph for your table.

(2)

Original question 27 (continued)

4MA1-2H Q13 M01, M02, M03, M04

MODULAR UNIT 2**6 marks**

- (c) Use your graph to find an estimate for the median weight of the 60 parcels.

..... kilograms
(1)

- (d) Use your graph to find an estimate for the number of these parcels that weigh more than 3.7 kilograms.

.....
(2)

Worked solution 27

MODULAR UNIT 2

Complete a cumulative-frequency table / Draw a cumulative-frequency graph / Read the median from the graph
/ Read the number above 3.7 kg

6 marks**M01 – Complete a cumulative-frequency table****Add frequencies successively**

$$4, 4 + 15 = 19, 19 + 20 = 39, 39 + 11 = 50, 50 + 6 = 56, 56 + 4 = 60.$$

Final answer

4, 19, 39, 50, 56, 60

M02 – Draw a cumulative-frequency graph**Plot upper-bound points**

$$(1, 4), (2, 19), (3, 39), (4, 50), (5, 56), (6, 60).$$

Join

Join the points with the accepted smooth curve or line segments.

Final answer

Correct cumulative-frequency graph

M03 – Read the median from the graph**Use the middle cumulative frequency**

Half of 60 is 30; reading across at CF=30 gives about 2.3–2.7 kg.

Final answer

2.3 to 2.7 kg

M04 – Read the number above 3.7 kg**Read CF at 3.7**

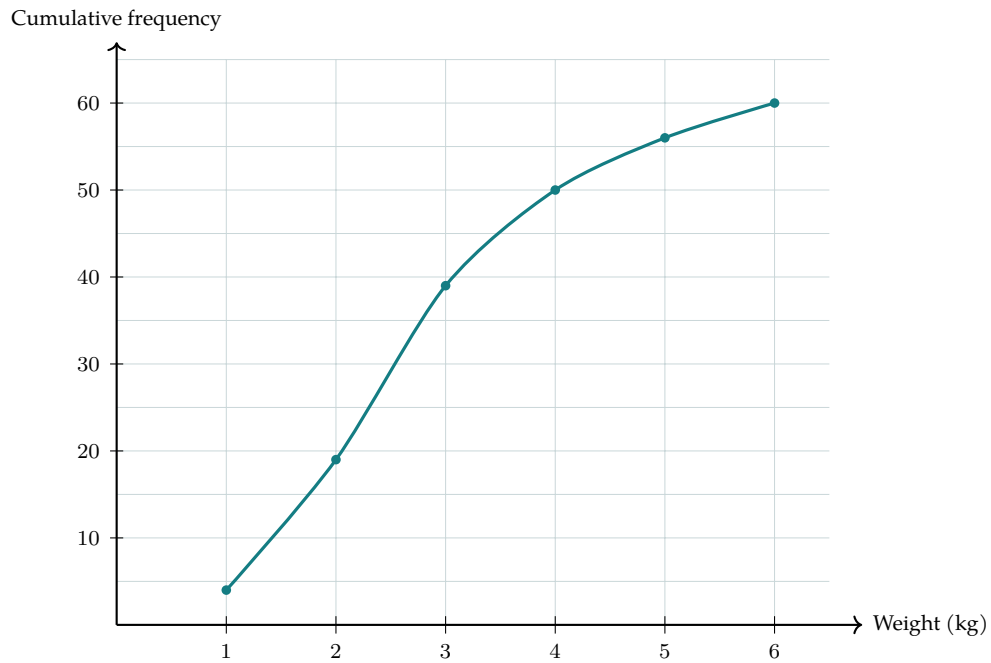
The graph gives approximately 45–48 parcels at or below 3.7 kg.

Subtract from 60

The number above 3.7 kg is 12–15.

Final answer

12 to 15 parcels

Completed solution diagram: Completed cumulative-frequency graph

Original question 26

4MA1-2H Q13 Q13(a), Q13(b), Q13(c), Q13(d)

ORIGINAL QUESTION**7 marks**

13 The table gives information about the ages, in years, of 80 people in a train carriage.

Age (a years)	Frequency
$0 < a \leq 20$	7
$20 < a \leq 30$	25
$30 < a \leq 40$	20
$40 < a \leq 50$	14
$50 < a \leq 60$	8
$60 < a \leq 70$	6

(a) Complete the cumulative frequency table.

Age (a years)	Cumulative frequency
$0 < a \leq 20$	
$0 < a \leq 30$	
$0 < a \leq 40$	
$0 < a \leq 50$	
$0 < a \leq 60$	
$0 < a \leq 70$	

(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.

Original question 26 (continued)

4MA1-2H Q13 Q13(a), Q13(b), Q13(c), Q13(d)

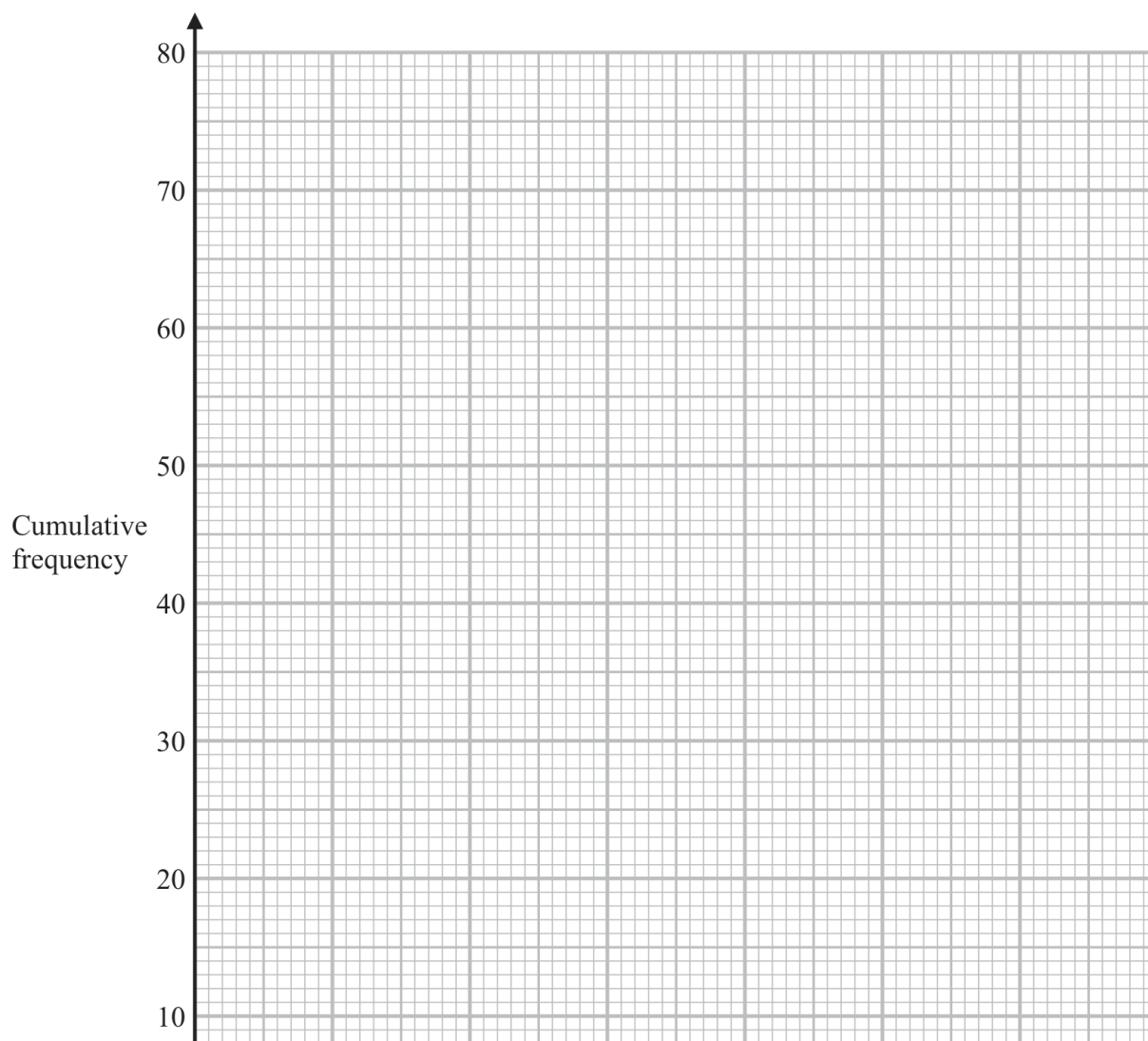
ORIGINAL QUESTION**7 marks**

(2)

(c) Use your graph to find an estimate for the median age of the 80 people.

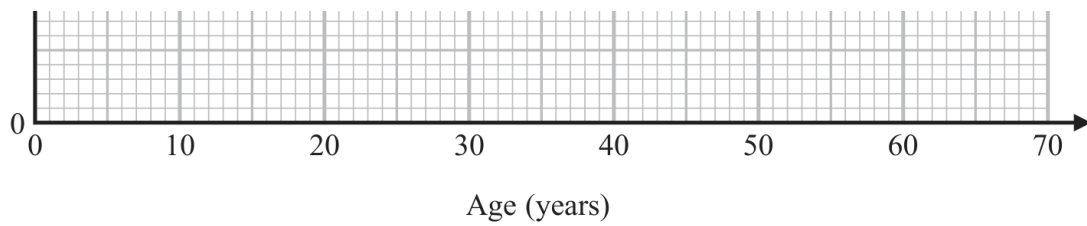
..... years

(1)



Original question 26 (continued)

4MA1-2H Q13 Q13(a), Q13(b), Q13(c), Q13(d)

ORIGINAL QUESTION**7 marks**

Of the people in the train carriage, 60% of those who are aged between 18 and 65 are going to work. None of the other people in the train carriage are going to work.

- (d) Use your graph to find an estimate for the number of people in the train carriage who are going to work.

.....
(3)

Worked solution 26

Chapter: Cumulative Frequency Diagrams

MODULAR UNIT 2

7 marks

Q13(a) – Chapter: Cumulative Frequency Diagrams

Family: Construct cumulative-frequency diagrams • Variant: Complete cumulative totals from a frequency table

Method

Successive cumulative totals are 7, 32, 52, 66, 74, 80.

*Why: Add each class frequency to the running total.***Final answer**

cumulative frequencies: 7, 32, 52, 66, 74, 80

Q13(b) – Chapter: Cumulative Frequency Diagrams

Family: Construct cumulative-frequency diagrams • Variant: Plot a cumulative-frequency curve from class frequencies

Method

Plot (20, 7), (30, 32), (40, 52), (50, 66), (60, 74), and (70, 80).

*Why: Cumulative frequencies are plotted at upper class boundaries.***Method**

Join the points with a smooth increasing curve or straight line segments.

*Why: A cumulative-frequency graph must preserve the cumulative order.***Final answer**

graph: CF points (20,7), (30,32), (40,52), (50,66), (60,74), (70,80)

Q13(c) – Chapter: Cumulative Frequency Diagrams

Family: Estimate and compare location and spread from cumulative frequency • Variant: Estimate the median from a cumulative-frequency diagram

Method

Read across from cumulative frequency 40 to the graph and then down to the age axis: the median is about 34 years.

*Why: The median is the 40th value out of 80.***Final answer**

median age: about 34 years

Q13(d) – Chapter: Cumulative Frequency Diagrams

Family: Read counts and thresholds from cumulative frequency • Variant: Estimate the number above or between thresholds

Method

From the graph, $CF(18)$ is about 6 and $CF(65)$ is about 77.

Why: Use cumulative-frequency readings at both age limits.

Method

About $77 - 6 = 71$ people are aged between 18 and 65.

Why: Subtract the lower cumulative count from the upper one.

Method

60% of 71 is 42.6, so an estimate is 43 people.

Why: Apply the percentage and give a whole-person estimate.

Final answer

people going to work: about 43 people

Completed diagram 26

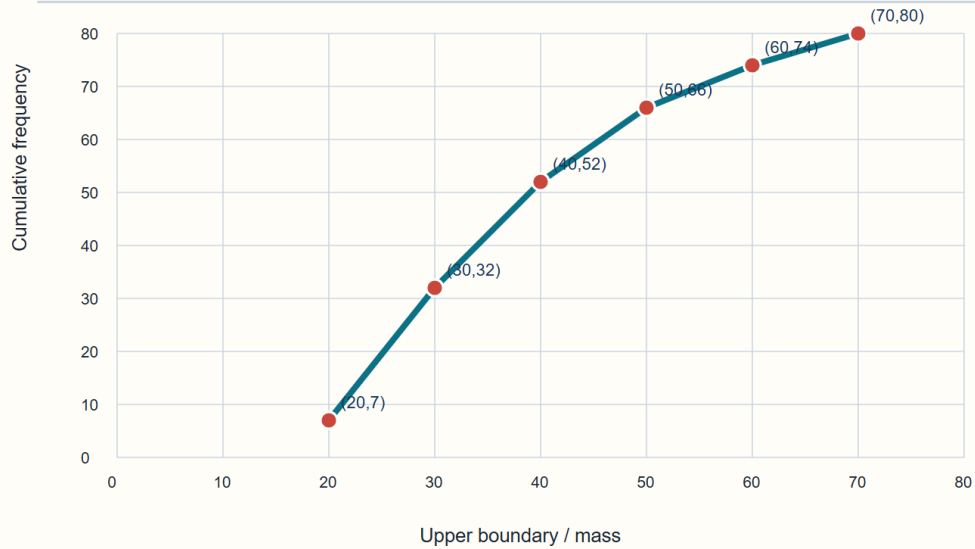
Chapter: Cumulative Frequency Diagrams

MODULAR UNIT 2

7 marks

Q13(b) Completed cumulative-frequency graph

Plot upper-bound points, then join in increasing order



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Original question 09

4MA1-1H Q12 Q12(a), Q12(b), Q12(c), Q12(d)

ORIGINAL QUESTION**6 marks**

- 12 The table gives information about the times, in minutes, taken by 80 customers to do their shopping in a supermarket.

Time taken (t minutes)	Frequency
$0 < t \leq 10$	7
$10 < t \leq 20$	26
$20 < t \leq 30$	24
$30 < t \leq 40$	14
$40 < t \leq 50$	7
$50 < t \leq 60$	2

- (a) Complete the cumulative frequency table.

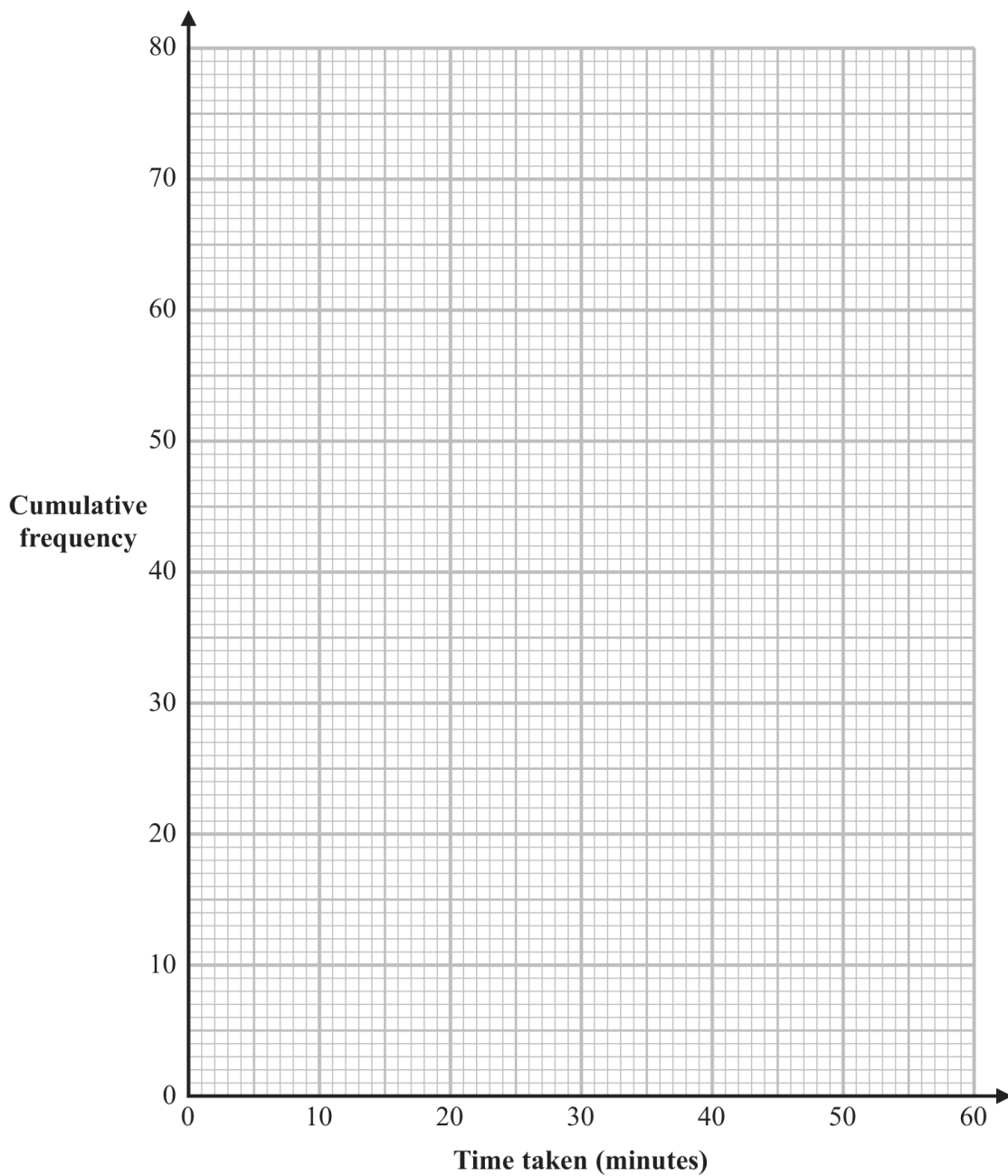
Time taken (t minutes)	Cumulative frequency
$0 < t \leq 10$	
$0 < t \leq 20$	
$0 < t \leq 30$	
$0 < t \leq 40$	
$0 < t \leq 50$	
$0 < t \leq 60$	

(1)

- (b) On the grid opposite, draw a cumulative frequency graph for your table.

Original question 09 (continued)

4MA1-1H Q12 Q12(a), Q12(b), Q12(c), Q12(d)

ORIGINAL QUESTION**6 marks**

(2)

Original question 09 (continued)

4MA1-1H Q12 Q12(a), Q12(b), Q12(c), Q12(d)

ORIGINAL QUESTION**6 marks**

(c) Use your graph to find an estimate for the median time taken.

..... minutes

(1)

One of the 80 customers is chosen at random.

(d) Use your graph to find an estimate for the probability that the time taken by this customer was more than 42 minutes.

.....

(2)

Worked solution 09

Chapter: Cumulative Frequency Diagrams

MODULAR UNIT 2

6 marks

Q12(a) – Chapter: Cumulative Frequency Diagrams

Family: Construct cumulative-frequency diagrams • Variant: Complete cumulative totals from a frequency table

Method

The cumulative frequencies are 7, 33, 57, 71, 78, 80.

*Why: Add the class frequencies successively.***Final answer**

cumulative frequencies: 7, 33, 57, 71, 78, 80

Q12(b) – Chapter: Cumulative Frequency Diagrams

Family: Construct cumulative-frequency diagrams • Variant: Plot a cumulative-frequency curve from class frequencies

Method

Plot (10,7), (20,33), (30,57), (40,71), (50,78), (60,80).

*Why: Cumulative frequency is plotted at the upper class boundaries.***Method**

Join the points with straight or smooth connected segments.

*Why: A cumulative-frequency graph joins the plotted points in order.***Final answer**

graph: Correct cumulative-frequency curve through the six points

Q12(c) – Chapter: Cumulative Frequency Diagrams

Family: Estimate and compare location and spread from cumulative frequency • Variant: Estimate the median from a cumulative-frequency diagram

Method

Half of 80 is 40; reading the cumulative-frequency curve gives a median time in the range 21–24 minutes.

*Why: Use $CF=40$ and follow the reviewed graph from part (b).***Final answer**

median: 21–24 minutes

Q12(d) – Chapter: Cumulative Frequency Diagrams

Family: Read counts and thresholds from cumulative frequency • Variant: Estimate a percentage from a cumulative curve

Method

The graph gives about 72 customers with time at most 42 minutes.

Why: Read the cumulative frequency at 42 minutes.

Method

Customers taking more than 42 minutes = $80 - 72 = 8$, so $P = 8/80 = 1/10$.

Why: Subtract from the total and divide by 80.

Final answer

probability: $8/80 = 1/10$

Completed diagram 09

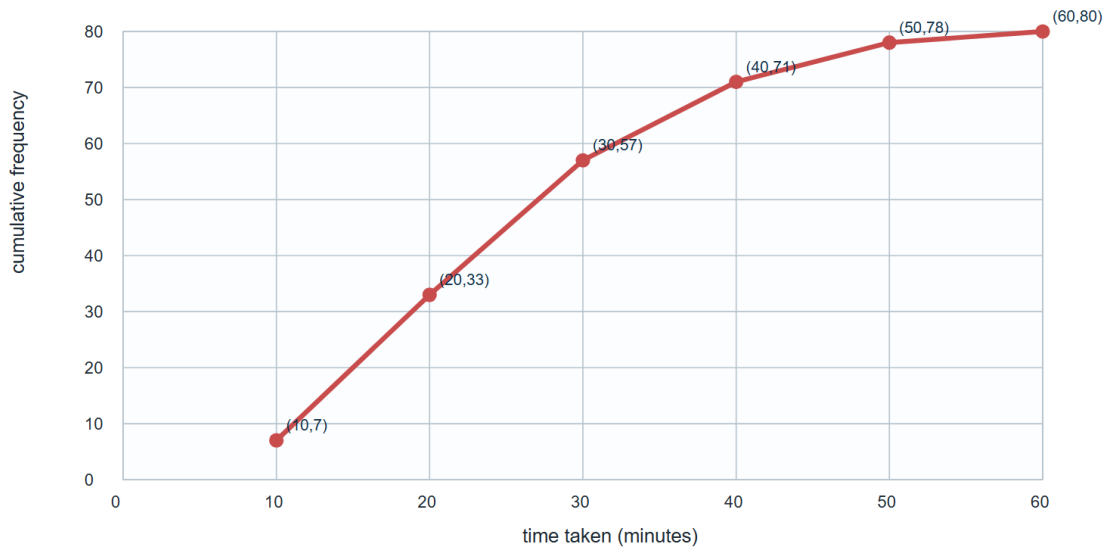
Chapter: Cumulative Frequency Diagrams

MODULAR UNIT 2

6 marks

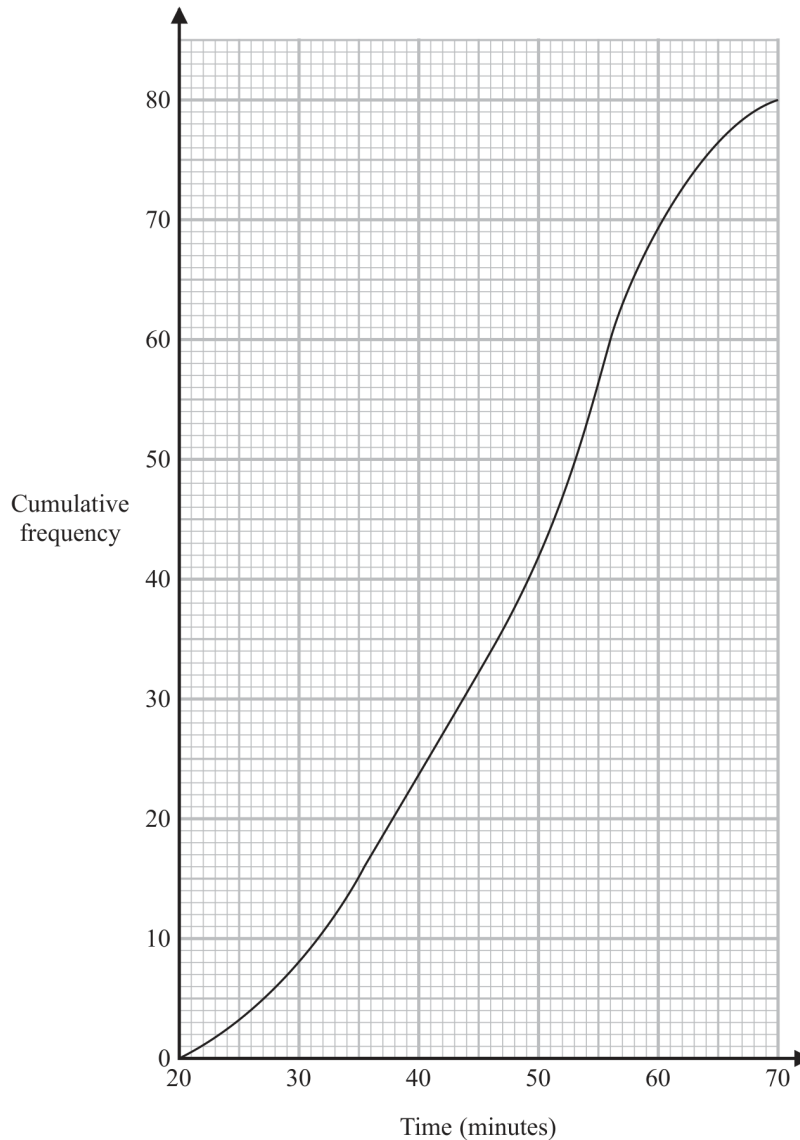
Q12 • Completed cumulative-frequency graph

Points at upper class boundaries



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- 12 A total of 80 men and women took part in a race.
The cumulative frequency graph gives information about the times, in minutes, they took for the race.



- (a) Use the graph to find an estimate for the interquartile range.

..... minutes
(2)

60% of the men took 50 minutes or less for the race.
No women took 50 minutes or less for the race.

- (b) Work out an estimate for the number of men who took part in the race.

.....
(3)

Worked solution

January 2020 | Question 12 | 2 marks

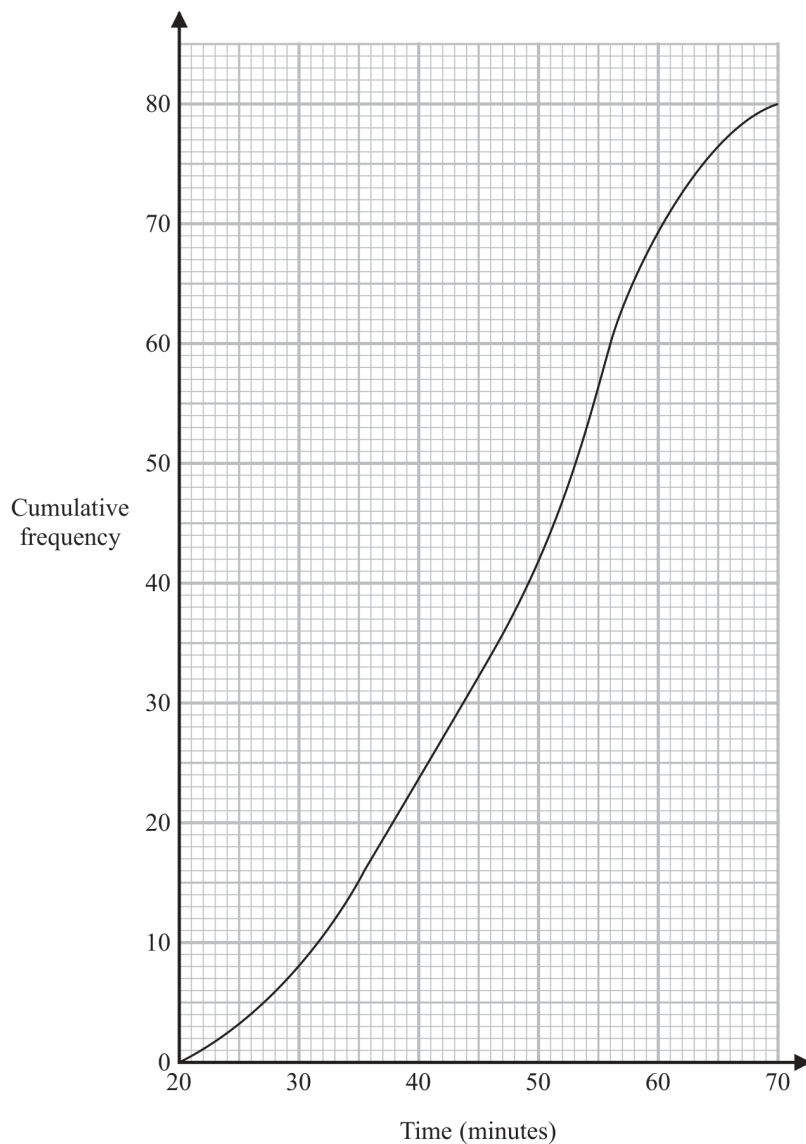
Family: Estimate and compare location and spread from cumulative frequency • Variant: Estimate IQR from a cumulative-frequency diagram

Method / working

1. Read Q_1 at $CF=20$ (about 38 min).
2. Read Q_3 at $CF=60$ (about 56 min), so $IQR=56-38=18$.

Final answer: about 18 minutes

- 12 A total of 80 men and women took part in a race.
The cumulative frequency graph gives information about the times, in minutes, they took for the race.



- (a) Use the graph to find an estimate for the interquartile range.

..... minutes
(2)

60% of the men took 50 minutes or less for the race.
No women took 50 minutes or less for the race.

- (b) Work out an estimate for the number of men who took part in the race.

.....
(3)

Worked solution

January 2020 | Question 12 | 3 marks

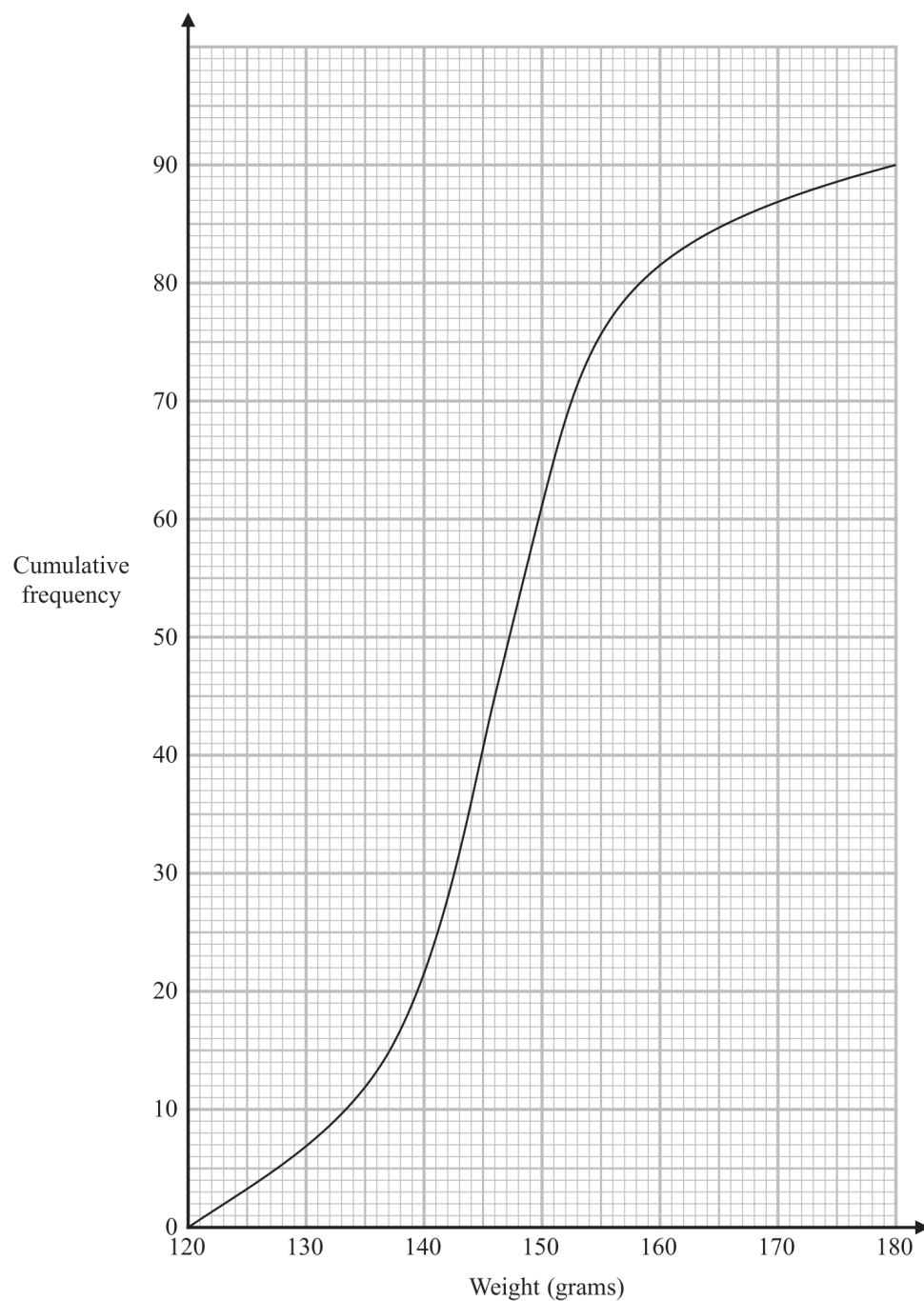
Family: Read counts and thresholds from cumulative frequency • Variant: Estimate a percentage from a cumulative curve

Method / working

1. At 50 minutes the graph gives about 42 people.
2. No women are at or below 50, so 42 is the number of men at or below 50.
3. $0.60 \times \text{men} = 42$, hence $\text{men} \sim 70$.

Final answer: about 70 men

- 11 The cumulative frequency graph gives information about the weights, in grams, of 90 bags of sweets.



- (a) Find an estimate for the median of the weights of these bags of sweets.

..... grams
(2)

Worked solution

January 2021 | Question 11 | 2 marks

Family: Estimate and compare location and spread from cumulative frequency • Variant: Estimate the median from a cumulative-frequency diagram

Method / working

1. Read the median at cumulative frequency 45.
2. The graph gives approximately 146 g (accepted range 145-147 g).

Final answer: 146 g

Roberto sells the bags of sweets to raise money for charity.

Bags with a weight greater than d grams are labelled large bags and sold for 3.75 euros each bag.

The total amount of money he receives by selling all the large bags is 93.75 euros.

(b) Find the value of d .

$d = \dots\dots\dots$
(3)

Worked solution

January 2021 | Question 11 | 3 marks

Family: Read counts and thresholds from cumulative frequency • Variant: Recover a threshold from a target count or amount

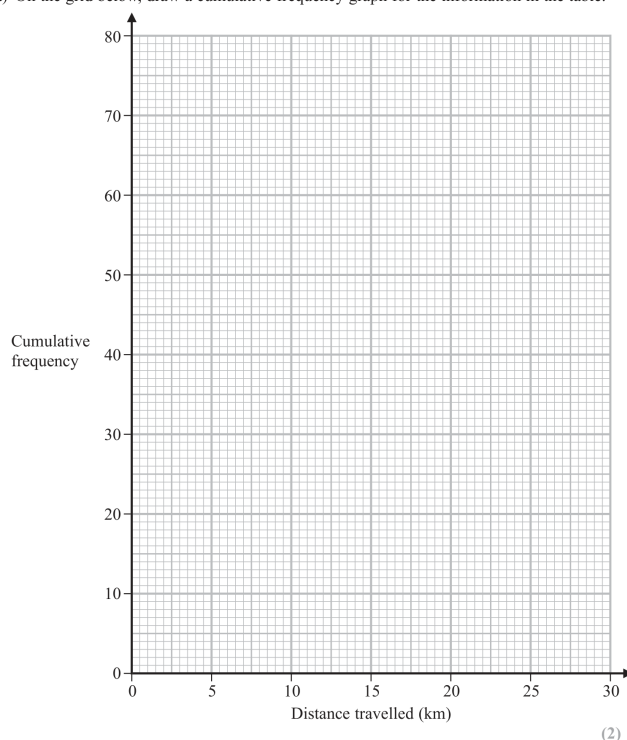
Method / working

1. Large-bag frequency = $93.75/3.75=25$.
 2. So cumulative frequency at d is $90-25=65$.
 3. Reading CF=65 gives d~151 g.
- Final answer: d=151

- 12 The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office A.

Distance travelled (d km)	Cumulative frequency
$0 < d \leq 5$	17
$0 < d \leq 10$	32
$0 < d \leq 15$	57
$0 < d \leq 20$	70
$0 < d \leq 25$	76
$0 < d \leq 30$	80

- (a) On the grid below, draw a cumulative frequency graph for the information in the table.



(2)

- (b) Use your graph to find an estimate for the median distance travelled.

..... km
(1)

- (c) Use your graph to find an estimate for the interquartile range of the distances travelled.

..... km
(2)

For Office B, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

- (d) Use the information above to compare the distances that workers at Office A and workers at Office B travel from home to work.
Write down **two** comparisons.

- 1
.....
.....
2
.....
.....
(2)

Worked solution

June 2021 | Question 12 | 2 marks

Family: Construct cumulative-frequency diagrams • Variant: Plot from an existing cumulative-frequency table

Method / working

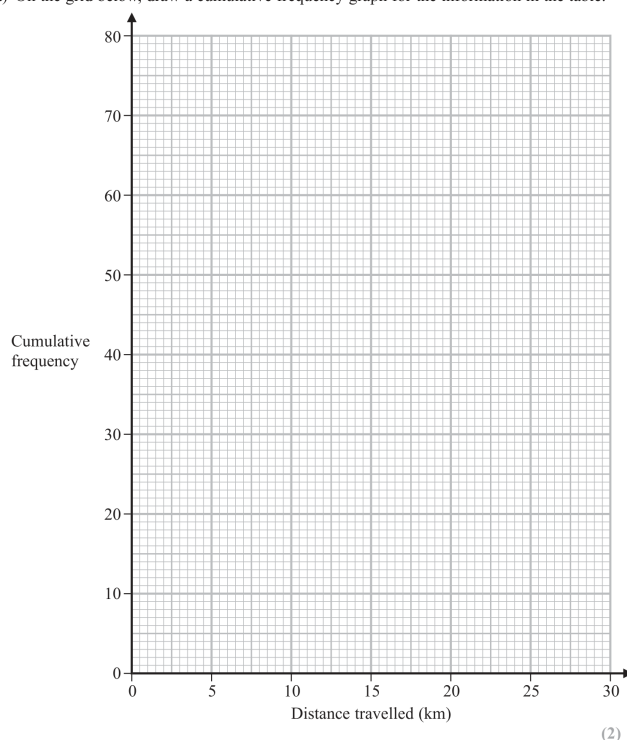
1. Plot the six cumulative-frequency points at the upper class boundaries (or accepted class midpoints).
2. Join them with a smooth increasing curve/line segments, not bars.

Final answer: Cumulative-frequency graph

- 12 The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office A.

Distance travelled (d km)	Cumulative frequency
$0 < d \leq 5$	17
$0 < d \leq 10$	32
$0 < d \leq 15$	57
$0 < d \leq 20$	70
$0 < d \leq 25$	76
$0 < d \leq 30$	80

- (a) On the grid below, draw a cumulative frequency graph for the information in the table.



(2)

- (b) Use your graph to find an estimate for the median distance travelled.

..... km
(1)

- (c) Use your graph to find an estimate for the interquartile range of the distances travelled.

..... km
(2)

For Office B, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

- (d) Use the information above to compare the distances that workers at Office A and workers at Office B travel from home to work.
Write down **two** comparisons.

- 1
-
- 2
-
-
- (2)

Worked solution

June 2021 | Question 12 | 1 marks

Family: Estimate and compare location and spread from cumulative frequency • Variant: Estimate the median from a cumulative-frequency diagram

Method / working

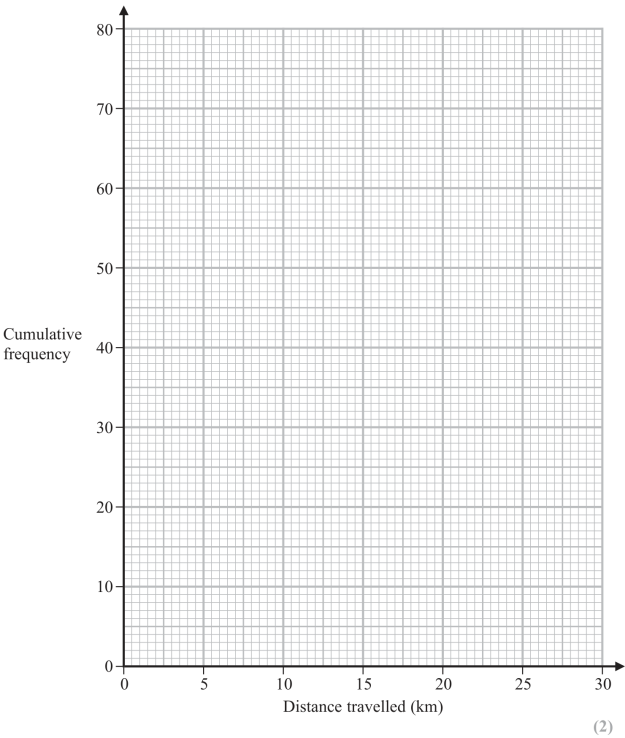
1. Read the graph at cumulative frequency 40 (the median position).
2. The estimate is in the accepted range 10.5-12 km.

Final answer: 10.5-12 km

12 The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office A.

Distance travelled (d km)	Cumulative frequency
$0 < d \leq 5$	17
$0 < d \leq 10$	32
$0 < d \leq 15$	57
$0 < d \leq 20$	70
$0 < d \leq 25$	76
$0 < d \leq 30$	80

(a) On the grid below, draw a cumulative frequency graph for the information in the table.



(b) Use your graph to find an estimate for the median distance travelled.

..... km
(1)

(c) Use your graph to find an estimate for the interquartile range of the distances travelled.

..... km
(2)

For Office B, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

(d) Use the information above to compare the distances that workers at Office A and workers at Office B travel from home to work.
Write down **two** comparisons.

- 1
-
-
- 2
-
-
-
- (2)

Worked solution

June 2021 | Question 12 | 2 marks

Family: Estimate and compare location and spread from cumulative frequency • Variant: Estimate IQR from a cumulative-frequency diagram

Method / working

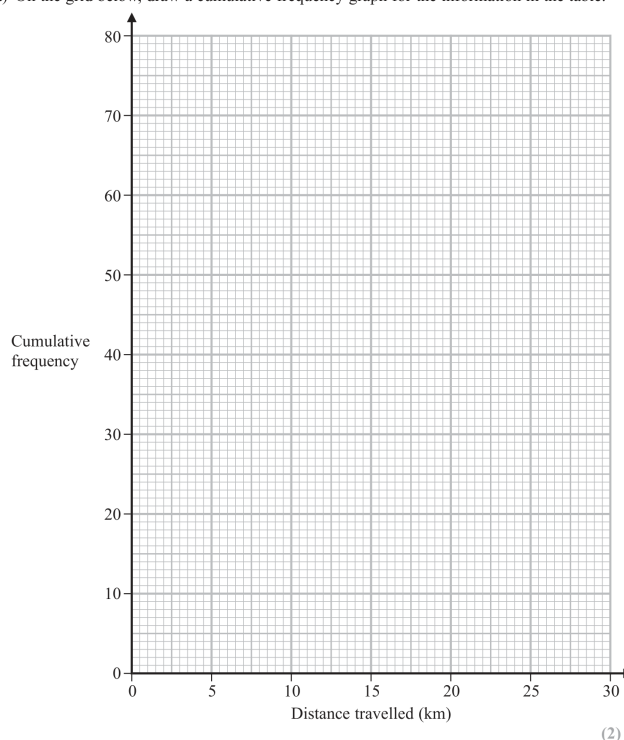
1. Read lower and upper quartiles at CF 20 and CF 60.
2. Subtract the two readings to obtain an IQR in the accepted range 8.5-11.5 km.

Final answer: 8.5-11.5 km

- 12 The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office *A*.

Distance travelled (<i>d</i> km)	Cumulative frequency
$0 < d \leq 5$	17
$0 < d \leq 10$	32
$0 < d \leq 15$	57
$0 < d \leq 20$	70
$0 < d \leq 25$	76
$0 < d \leq 30$	80

- (a) On the grid below, draw a cumulative frequency graph for the information in the table.



(2)

- (b) Use your graph to find an estimate for the median distance travelled.

..... km
(1)

- (c) Use your graph to find an estimate for the interquartile range of the distances travelled.

..... km
(2)

For Office *B*, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

- (d) Use the information above to compare the distances that workers at Office *A* and workers at Office *B* travel from home to work.
Write down **two** comparisons.

- 1
.....
.....
2
.....
.....
(2)

Worked solution

June 2021 | Question 12 | 2 marks

Family: Estimate and compare location and spread from cumulative frequency • Variant: Compare two distributions using cumulative-frequency measures

Method / working

1. Compare medians: Office B median 15 km is higher, so B workers tend to travel further.
2. Compare IQRs: Office A's IQR is larger than B's 5 km, so A distances are more varied.

Final answer: Office B travels further; Office A is more spread out

- 11 The table gives information about the times taken by 90 runners to complete a 10km race.

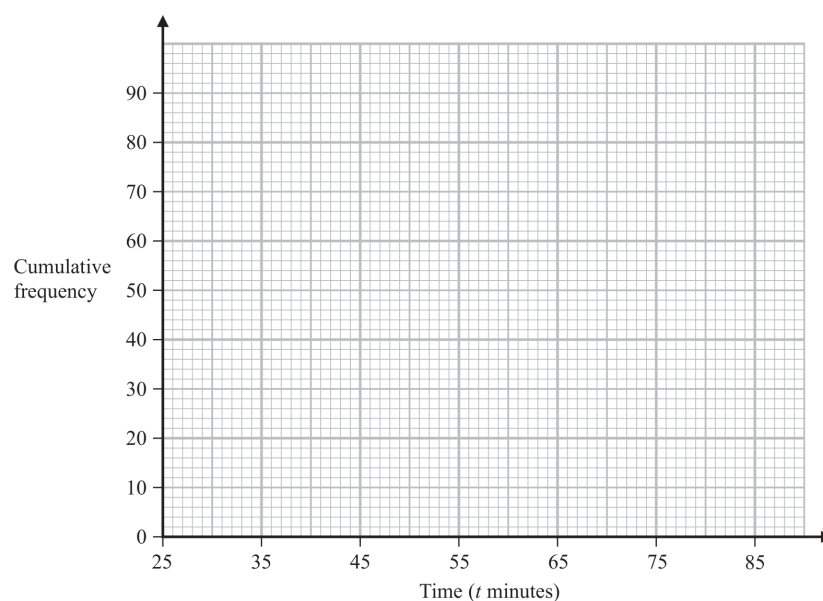
Time (t minutes)	Frequency
$25 < t \leq 35$	12
$35 < t \leq 45$	24
$45 < t \leq 55$	28
$55 < t \leq 65$	12
$65 < t \leq 75$	10
$75 < t \leq 85$	4

- (a) Complete the cumulative frequency table.

Time (t minutes)	Cumulative frequency
$25 < t \leq 35$	12
$25 < t \leq 45$	
$25 < t \leq 55$	
$25 < t \leq 65$	
$25 < t \leq 75$	
$25 < t \leq 85$	

(1)

- (b) On the grid below, draw a cumulative frequency graph for your table.



(2)

Any runner who completed the race in a time T minutes such that $42 < T \leq 52$ minutes was awarded a silver medal.

- (c) Use your graph to find an estimate for the number of runners who were awarded a silver medal.

..... runners
(2)

Worked solution

November 2021 | Question 11 | 1 marks

Family: Construct cumulative-frequency diagrams | Variant: Complete cumulative totals from a frequency table

METHOD / WORKING

1. The cumulative frequencies are

2. 12, 36, 64, 76, 86, 90

3. For $42 < T \leq 52$,

$$4. F(52) \approx 36 + \frac{7}{10}(64 - 36) = 55.6$$

$$5. F(42) \approx 12 + \frac{7}{10}(36 - 12) = 28.8$$

$$6. 55.6 - 28.8 = 26.8$$

7. Answer: about 27 runners.

FINAL ANSWER

about 27 runners.

- 11 The table gives information about the times taken by 90 runners to complete a 10km race.

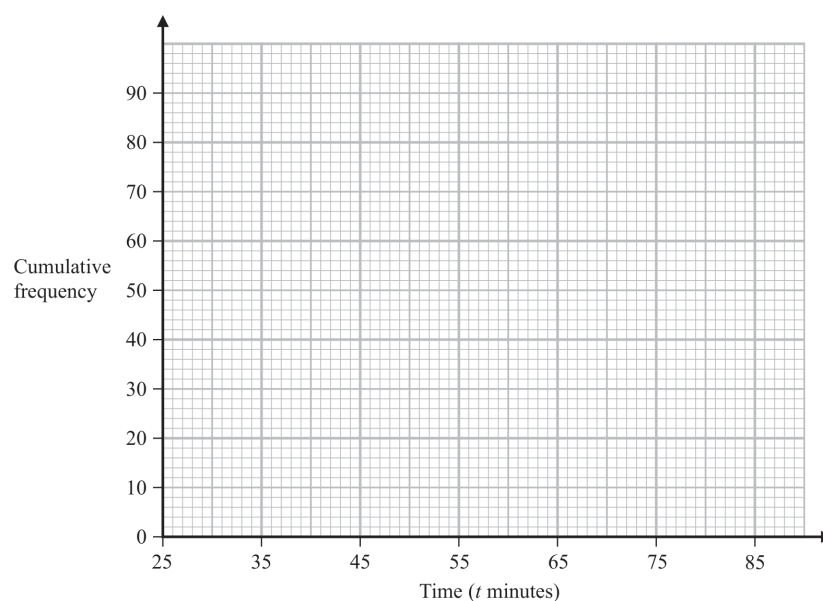
Time (t minutes)	Frequency
$25 < t \leq 35$	12
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$45 < t \leq 55$	28
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$65 < t \leq 75$	10
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$25 < t \leq 35$	12
$25 < t \leq 45$	
$25 < t \leq 55$	
$25 < t \leq 65$	
$25 < t \leq 75$	
$25 < t \leq 85$	

(1)

- (b) On the grid below, draw a cumulative frequency graph for your table.



(2)

Any runner who completed the race in a time T minutes such that $42 < T \leq 52$ minutes was awarded a silver medal.

- (c) Use your graph to find an estimate for the number of runners who were awarded a silver medal.

..... runners
(2)

Worked solution

November 2021 | Question 11 | 2 marks

Family: Construct cumulative-frequency diagrams | Variant: Plot a cumulative-frequency curve from class frequencies

METHOD / WORKING

1. The cumulative frequencies are

2. 12, 36, 64, 76, 86, 90

3. For $42 < T \leq 52$,

$$4. F(52) \approx 36 + \frac{7}{10}(64 - 36) = 55.6$$

$$5. F(42) \approx 12 + \frac{7}{10}(36 - 12) = 28.8$$

$$6. 55.6 - 28.8 = 26.8$$

7. Answer: about 27 runners.

FINAL ANSWER

about 27 runners.

- 11 The table gives information about the times taken by 90 runners to complete a 10km race.

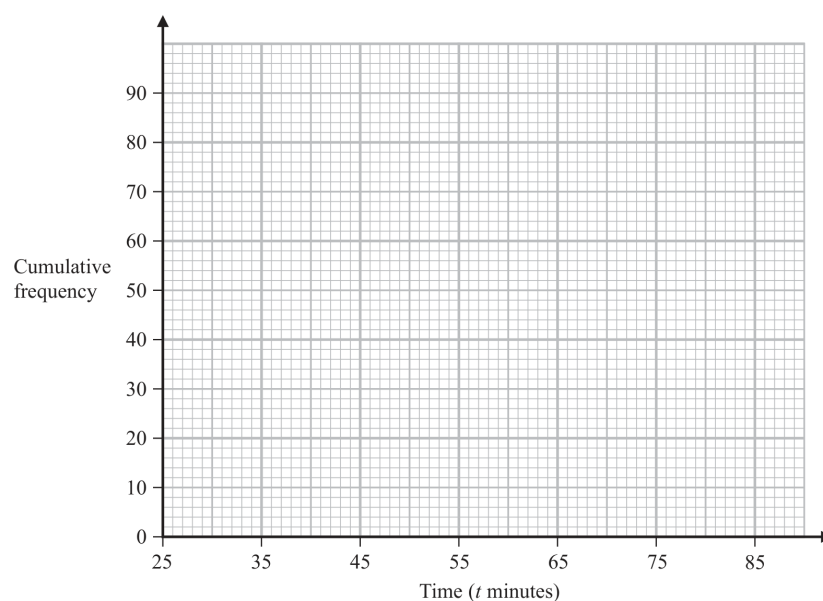
Time (t minutes)	Frequency
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- (a) Complete the cumulative frequency table.

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$25 < t \leq 35$	12
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$25 < t \leq 55$	
$25 < t \leq 65$	
$25 < t \leq 75$	
$25 < t \leq 85$	

(1)

- (b) On the grid below, draw a cumulative frequency graph for your table.



(2)

Any runner who completed the race in a time T minutes such that $42 < T \leq 52$ minutes was awarded a silver medal.

- (c) Use your graph to find an estimate for the number of runners who were awarded a silver medal.

..... runners
(2)

Worked solution

November 2021 | Question 11 | 2 marks

Family: Read counts and thresholds from cumulative frequency | Variant: Estimate the number above or between thresholds

METHOD / WORKING

1. The cumulative frequencies are

2. 12, 36, 64, 76, 86, 90

3. For $42 < T \leq 52$,

$$4. F(52) \approx 36 + \frac{7}{10}(64 - 36) = 55.6$$

$$5. F(42) \approx 12 + \frac{7}{10}(36 - 12) = 28.8$$

$$6. 55.6 - 28.8 = 26.8$$

7. Answer: about 27 runners.

FINAL ANSWER

about 27 runners.

10 The table shows information about the number of minutes each of 120 buses was late last Monday.

Number of minutes late (L)	Frequency
$0 < L \leq 10$	10
$10 < L \leq 20$	16
$20 < L \leq 30$	44
$30 < L \leq 40$	29
$40 < L \leq 50$	15
$50 < L \leq 60$	6

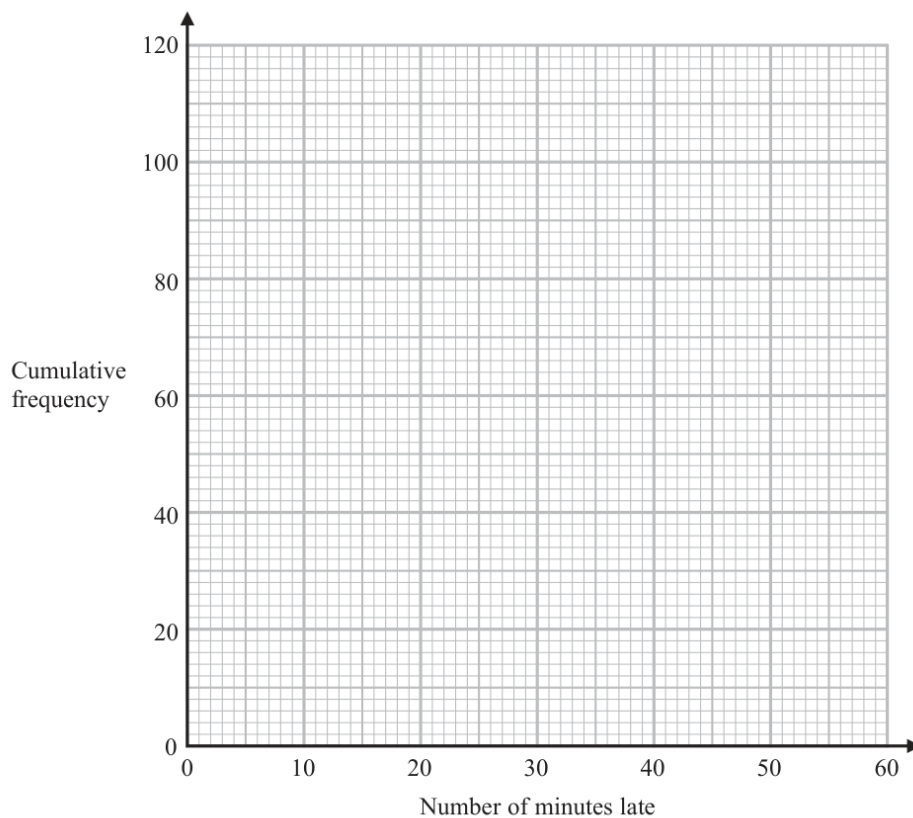
(a) Complete the cumulative frequency table below.

Number of minutes late (L)	Cumulative frequency
$0 < L \leq 10$	
$0 < L \leq 20$	
$0 < L \leq 30$	
$0 < L \leq 40$	
$0 < L \leq 50$	
$0 < L \leq 60$	

(1)



(b) On the grid, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the interquartile range.

.....minutes
(2)

(d) Use your graph to find an estimate for the number of buses that were more than 48 minutes late last Monday.

.....
(2)

(Total for Question 10 is 7 marks)

Answer reference | May-Nov 2020 | Question 10

Family: Construct cumulative-frequency diagrams • Variant: Complete cumulative totals from a frequency table

10	a		10, 26, 70, 99, 114, 120	1	B1
	b		correct cumulative frequency graph	2	B2 fully correct cf graph – points at ends of intervals and joined with curve or line segments If not B2 then B1 for 5 or 6 (ft from a table with only one arithmetic error) of their points at ends of intervals and joined with curve or line segments OR for 5 or 6 points plotted correctly at ends of intervals not joined OR for 5 or 6 of their points from table plotted consistently within each interval (not at upper ends of intervals) at their correct heights and joined with smooth curve or line segments
	c				M1 For use of 30 and 90, or 30.25 and 90.75 (eg reading of 21 and 37 stated or indicated by marks on horizontal axis that correspond to 30 (or 30.25) and 90 (or 90.75) on the vertical axis or correct readings ft their cf graph provided method to show readings is shown)
			16	2	A1 accept 14 – 18, ft from their cf graph (ft provided method to show readings is shown)
	d				M1 For use of cf from number of minutes late being 48 (eg an indication by a mark on the vertical axis corresponding to 48 mins late or a correct reading ft their cf graph)
			9	2	A1 accept 7 – 10, ft from their cf graph
Total 7 marks					

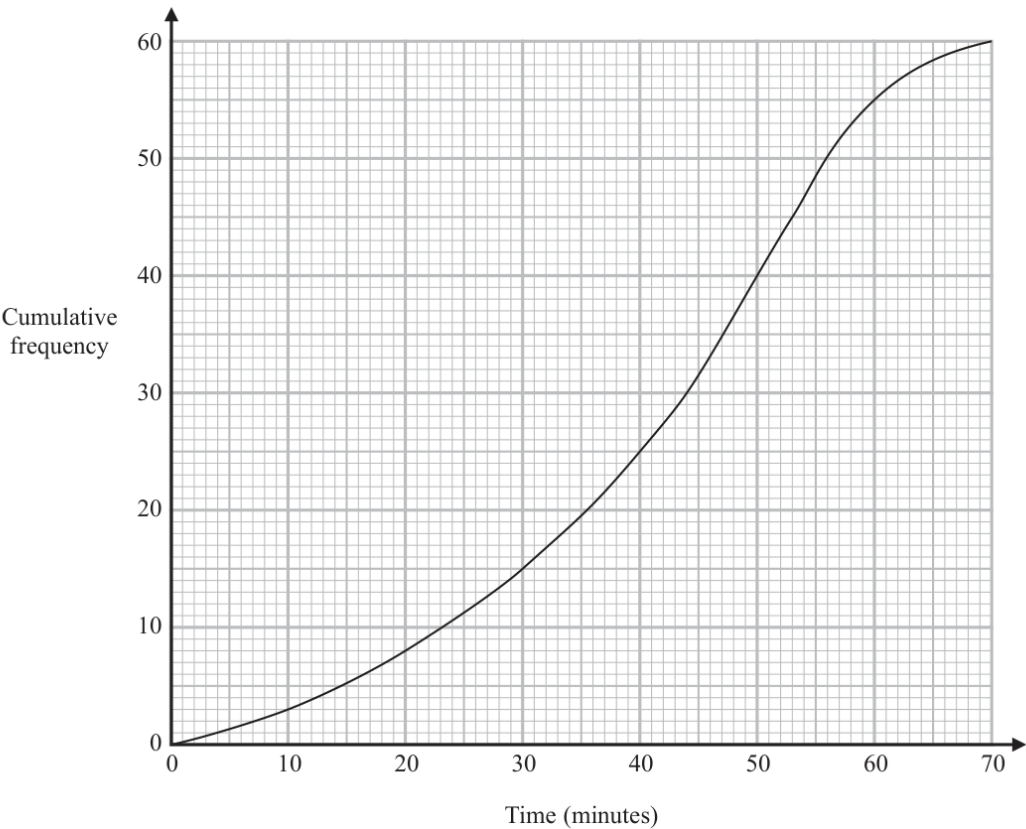
Worked solution

May-Nov 2020 | Question 10 | 7 marks

Family: Construct cumulative-frequency diagrams • Variant: Complete cumulative totals from a frequency table

Official final mark-scheme pages are embedded immediately after this question.

12 The cumulative frequency graph gives information about the time, in minutes, each of 60 people took to shop in a market.



(a) Use the graph to find an estimate for the median time people took to shop in the market.

..... minutes
(1)

(b) Use the graph to find an estimate for the number of people who took longer than 55 minutes to shop in the market.

.....
(2)



- (c) Use the graph to complete the frequency table to give information about the time, in minutes, each of the 60 people took to shop in the market.

Time taken to shop in the market (m minutes)	Frequency
$0 < m \leq 10$	3
$10 < m \leq 20$	5
$20 < m \leq 30$	
$30 < m \leq 40$	
$40 < m \leq 50$	
$50 < m \leq 60$	
$60 < m \leq 70$	

(2)

(Total for Question 12 is 5 marks)

Answer reference | January 2023 | Question 12

Family: Read counts and thresholds from cumulative frequency • Variant: Estimate the number above or between thresholds

Question	working	Answer	Mark	Notes																
12 (a)		43.5 - 44.5	1	B1 ±0.5 small square																
(b)	eg reading of 48 - 49		2	M1 For correct method to start the question eg a vertical line from 55 up to the line and a horizontal line from the correct point on the curve or a mark on the curve at the correct point and a mark on the vertical axis at the correct point or a correct reading of 48 to 49																
	Correct answer scores full marks (unless from obvious incorrect working)	11 or 12		A1 Allow an answer of 11 or 12 (ie must be whole number)																
(c)	<table><tr><th>Time taken to shop in the market (m minutes)</th><th>Frequency</th></tr><tr><td>$0 < m \leq 10$</td><td>3</td></tr><tr><td>$10 < m \leq 20$</td><td>5</td></tr><tr><td>$20 < m \leq 30$</td><td>7</td></tr><tr><td>$30 < m \leq 40$</td><td>10</td></tr><tr><td>$40 < m \leq 50$</td><td>15</td></tr><tr><td>$50 < m \leq 60$</td><td>15</td></tr><tr><td>$60 < m \leq 70$</td><td>5</td></tr></table>	Time taken to shop in the market (m minutes)	Frequency	$0 < m \leq 10$	3	$10 < m \leq 20$	5	$20 < m \leq 30$	7	$30 < m \leq 40$	10	$40 < m \leq 50$	15	$50 < m \leq 60$	15	$60 < m \leq 70$	5		2	B2 All values correctly filled in (NB: first 2 are already completed) (B1 for 3 or 4 correct values from 7, 10, 15, 15, 5)
Time taken to shop in the market (m minutes)	Frequency																			
$0 < m \leq 10$	3																			
$10 < m \leq 20$	5																			
$20 < m \leq 30$	7																			
$30 < m \leq 40$	10																			
$40 < m \leq 50$	15																			
$50 < m \leq 60$	15																			
$60 < m \leq 70$	5																			
				Total 5 marks																

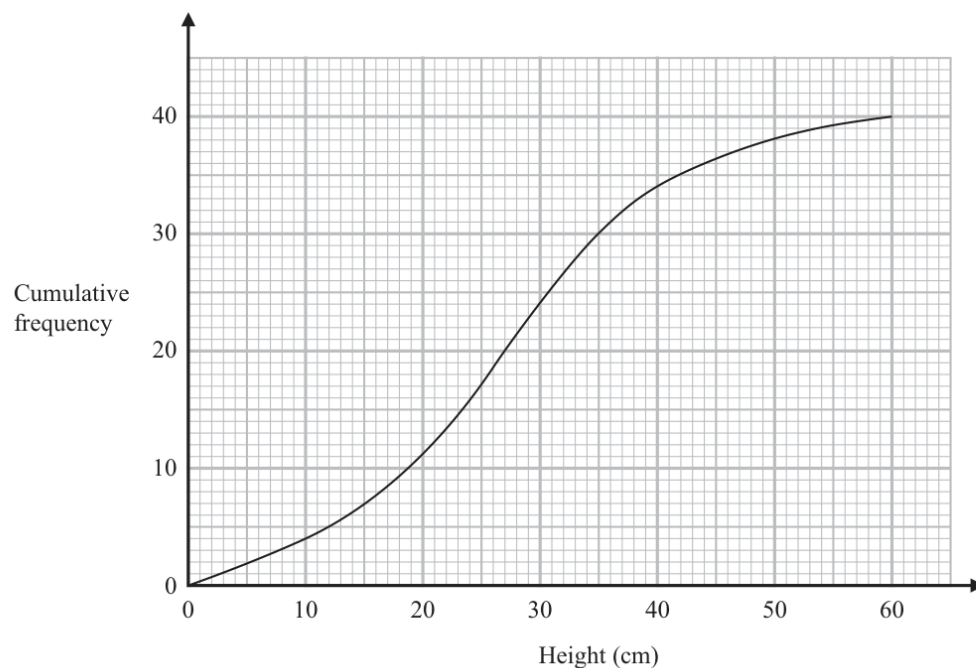
Worked solution

January 2023 | Question 12 | 5 marks

Family: Read counts and thresholds from cumulative frequency • Variant: Estimate the number above or between thresholds

Official final mark-scheme pages are embedded immediately after this question.

- 14 The cumulative frequency graph shows information about the heights of 40 plants that Greta has grown.



- (a) Use the graph to find an estimate for the median height.

..... cm
(1)

- (b) Use the graph to find an estimate for the interquartile range of the heights.

..... cm
(2)



Plants with a height greater than 40 cm are premium plants.
Greta sells all the premium plants for 30 euros each.

(c) Work out the total amount of money Greta receives for the premium plants.

..... euros
(2)

(Total for Question 14 is 5 marks)

Worked solution

November 2025 | Question 14 | 5 marks

Family: Estimate and compare location and spread from cumulative frequency | Variant: Estimate lower and upper quartiles from cumulative frequency

METHOD / WORKING

Worked solution
November 2025 Question 14 5 marks
Family: Estimate and compare location and spread from cumulative frequency Variant: Estimate lower and upper quartiles from cumulative frequency
Solution
There are 40 plants.
From the graph,
medianapprox 26 cm
$Q_1\approx 19\text{cm}, Q_3\approx 35\text{cm}$
$IQR\approx 35-19=16\text{ cm}$
For premium plants, use height $>40\text{ cm}$.
$F(40)\approx 34$
$40-34=6$
$6 \times 30=180$
Answers: median about 26 cm, IQR about 16 cm, and 180 euros.
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER